INTERNATIONAL SEARCH REPORT

International application No.
PCT/CN2025/076960

A. CLASSIFICATION OF SUBJECT MATTER

G01S11/02(2010.01)i; H04L1/00(2006.01)i; H04W4/02(2018.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

IPC: G01S, H04L, H04W

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

CNTXT, VEN, DWPI, ENTXT, CNKI: UWB, 超宽带, 测距, 接收, 数据帧, 信息帧, 时隙, 结束, 失败, 转发, Final data, ultra wide band, rang+, receiv+, frame, time slot, end, finish, failure, transimt+, retransmission, transpond.

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	CN 101990244 A (ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE) 23 March 2011 (2011-03-23) description, paragraphs 44-46	1-12
A	CN 117461337 A (SAMSUNG ELECTRONICS CO., LTD.) 26 January 2024 (2024-01-26) entire document	1-12
A	CN 117641234 A (APPLE INC.) 01 March 2024 (2024-03-01) entire document	1-12
A	CN 114730003 A (SAMSUNG ELECTRONICS CO., LTD.) 08 July 2022 (2022-07-08) entire document	1-12
A	WO 2012163136 A1 (ZTE CORP. et al.) 06 December 2012 (2012-12-06) entire document	1-12
A	KR 20230078924 A (SAMSUNG ELECTRONICS CO., LTD.) 05 June 2023 (2023-06-05) entire document	1-12

☐ Further documents are listed in the continuation of Box C. ☑ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
30 May 2025	**03 June 2025**

Name and mailing address of the ISA/CN	Authorized officer
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INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/CN2025/076960

Patent document cited in search report	Publication date (day/month/year)	Patent family member(s)	Publication date (day/month/year)
CN 101990244 A	23 March 2011	KR 20110014042 A	10 February 2011
		KR 101192430 B1	17 October 2012
		US 2011034195 A1	10 February 2011
CN 117461337 A	26 January 2024	WO 2022260497 A1	15 December 2022
		EP 4342203 A1	27 March 2024
		EP 4342203 A4	07 August 2024
		US 2022397659 A1	15 December 2022
CN 117641234 A	01 March 2024	None	
CN 114730003 A	08 July 2022	KR 20220104187 A	26 July 2022
		EP 4024081 A1	06 July 2022
		EP 4024081 A4	19 October 2022
		EP 4024081 B1	30 October 2024
		US 2021173064 A1	10 June 2021
		US 11965951 B2	23 April 2024
		WO 2021112380 A1	10 June 2021
WO 2012163136 A1	06 December 2012	None	
KR 20230078924 A	05 June 2023	None	